Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Python · Distributed Systems · REST APIs · AWS · CI/CD · AI/ML Integration · Full-Stack · Production Engineering · Mentoring

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8 years delivering production Python systems in complex, compliance-critical SaaS environments - and a track record of integrating AI and machine learning to enhance real software capabilities. At Entrata, led end-to-end delivery of 20+ production modules across 120+ Agile sprints: Python and JavaScript backends, event-driven async architecture, cloud-native AWS infrastructure, and CI/CD-driven quality at 99.5% uptime for 100K+ monthly active users. Conducted 300+ code reviews and mentored 5+ engineers. Built and shipped two AI systems from scratch: AgentProbe (open-source LLM evaluation platform with CI-integrated regression gating) and LeaseI (8-agent agentic AI system with RAG pipelines, HITL escalation, and 93.6% precision across 1,240 test cases). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & Backend: Python (8yr), JavaScript, TypeScript, SQL, PHP, Bash, FastAPI, Flask, REST APIs, Microservices, Object-Oriented Design, Docker, CI/CD (GitHub Actions), Git, Linux

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), RabbitMQ (event-driven async, retry, idempotency), Redis, PostgreSQL, MySQL, Oracle, NoSQL, Apache Spark, PySpark, ETL, Data Modeling, Data Pipelines, Distributed Systems, Observability, Performance Engineering

AI & ML Integration: LangGraph, LangChain, Agentic AI, RAG Pipelines, LLM Evaluation, LLM-as-Judge, HITL, Guardrails, Prompt Engineering, ChromaDB, Vector Search, Embeddings, OpenAI API, Anthropic API, PyTorch, scikit-learn, BERT, spaCy, NLP, NumPy, Pandas, Tableau, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built and deployed a production agentic AI system (LangGraph, FastAPI, Python) - iterative development from requirements through CI/CD deployment; deterministic guardrails on outputs; automated test suite gating every release.
- Translated complex business requirements into scalable, maintainable software; observability on system performance, latency, and reliability.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built Python ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end from data ingestion through model evaluation and statistical validation across 190+ countries.
- Mentored 600+ students in Python, SQL, OOP, and software engineering fundamentals; designed monitoring dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **End-to-end ownership of business-critical production systems** - 20+ scalable Python and JavaScript modules across 120+ Agile sprints; 99.5% uptime SLA at 100K+ monthly active users; translated complex compliance and business requirements into measurable technical outcomes across a multi-tenant SaaS platform.
- **Distributed systems and cloud-native infrastructure** - RabbitMQ event-driven async processing with retry, idempotency, and dead-letter guarantees; Redis caching; Docker containerization (50% scalability improvement, 30% deployment reduction); AWS S3, EC2, RDS, Glue at enterprise scale; owned HUD income-limit data pipeline 7 years (federal source → 180,000 domain records powering eligibility for 100K+ housing units).
- **Code quality, performance, and reliability engineering** - 300+ peer code reviews; 200+ SQL/PostgreSQL/Oracle query optimizations (30% performance gain); reduced bug count to sub-10 for 6+ consecutive months; AI-driven observability tooling reducing triage time 40%; 100% PHI compliance with Libsodium-encrypted pipelines; GitHub Actions CI/CD.
- **Technical leadership and mentoring** - mentored 5+ engineers on Python, distributed systems, and engineering best practices; collaborated with stakeholders across product, compliance, and operations to define and deliver reliable technical solutions; participated in architecture and solution design reviews.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: Open-Source LLM Evaluation Platform AI/ML · CI/CD

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic · OpenAI | github.com/venkatalakshmiparimala/AgentProbe

Production AI quality platform: 15-endpoint FastAPI service, 5-metric evaluation engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate, LLM-as-judge scoring, multi-provider LLM routing. Zero regressions shipped across 1,240+ test cases.

Leasel: 8-Agent Agentic AI System with RAG Agentic · RAG · Production

Python · LangGraph · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production agentic system: 8-agent LangGraph stateful pipeline with multi-source RAG retrieval, deterministic guardrails on regulated outputs, HITL escalation, and source-linked answers. 93.6% precision, 87.9% F1 across 1,240 compliance test cases; CI quality gate with full observability.

AI on FHIR: NLP-Driven Healthcare Query System NLP · Full-Stack

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI application translating natural-language queries into FHIR API calls via BERT embeddings and LLM inference - reduced retrieval time ~60%; HIPAA-compliant RBAC with full data isolation across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017